

Pendulum Motion

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Abstract

A report on how physical factors impact the motion of a pendulum, specifically a pendulum consisting of a small mass attached to a massless rod and with subjection to a vertical driving force and gravity. This report makes use of the Lagrangian formalism and Python in order to understand how changes in initial conditions, such as angular frequency and initial angular displacement, affect the pendulum's motion. The report will show us the variety in the motion of pendulums, ranging from simple harmonic motion to stable inverted pendulum motion.

1 Introduction

In this report, we aim to understand the motion of pendulums consisting of a small mass attached to a massless rod, subjected to a vertical driving force. In some cases, the pendulum will be driven; in others, it will not. We will achieve this by first introducing and applying the Lagrangian formalism, and then solving and plotting the system's equation of motion using Python, based on our knowledge from [1]. The results will reveal the effects of initial conditions and driving factors on the pendulum motion, complimenting our theory work with the Python-produced results.

2 Main equations

To understand the motion of a system with one degree of freedom, we apply the Lagrangian formalism (taken from [1] and [2]). The system has one degree of freedom so its state can be described by a single generalized coordinate, q , and its time derivative, $\dot{q}$. The Lagrangian, $\mathcal{L}(q, \dot{q}, t)$, is defined as the difference between the system's kinetic energy, T , and potential energy, U :

$$\mathcal{L}(q, \dot{q}, t) = T - U \quad (1)$$

The equation of motion for the system can be derived using the Euler-Lagrange equation:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = 0 \quad (2)$$

This approach will provide us with the tools to investigate our pendulum systems.

2.1 The Driven Pendulum.

To derive the equation of motion for the vertically driven pendulum, we begin by expressing the coordinates of the pendulum bob in terms of the angular displacement, ϕ :

$$x = L \sin(\phi) \quad y = L \cos(\phi) + a \cos(\omega t) \quad (3)$$

where L is the pendulum length, a is the driving amplitude, and ω is the angular frequency. We can then write the velocities of the pendulum by taking time derivatives of the coordinates:

$$\dot{x} = L\dot{\phi}\cos(\phi) \quad \dot{y} = -L\dot{\phi}\sin(\phi) - a\omega\sin(\omega t) \quad (4)$$

Where a dot above a symbol represents a time derivative for example $\dot{x} = \frac{dx}{dt}$. Using the Lagrangian formalism Eq. 1 in combination with the formulas $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$ and $U = -mgy$ (taken from [2]), we can write the Lagrangian for the system as:

$$\mathcal{L}(\phi, \dot{\phi}, t) = T - U = \frac{1}{2}mL^2\dot{\phi}^2 + mgl\cos(\phi) + mL\omega^2\cos(\omega t)\cos(\phi) \quad (5)$$

We have omitted certain terms because they are total time derivatives of functions of coordinates and time, hence do not contribute to the equation of motion (a result found in [1] and [2]). Applying the Euler-Lagrange equation (Eq. 2) to this, after taking partial and time derivatives we obtain the equation of motion:

$$\ddot{\phi} = -\frac{g}{L}\sin(\phi) - \frac{a\omega^2}{L}\cos(\omega t)\sin(\phi) \quad (6)$$

By introducing the notation $\zeta = \frac{a}{L}$ and $\omega_0 = \sqrt{\frac{g}{L}}$, we can rewrite the equation of motion in a more readable format:

$$\ddot{\phi} = -(\omega_0^2 + \zeta\omega^2\cos(\omega t))\sin(\phi) \quad (7)$$

By choosing appropriate units of time, we can set $\omega_0 = 1$, so that Eq 7 simplifies to:

$$\ddot{\phi} = -(1 + \zeta\omega^2\cos(\omega t))\sin(\phi) \quad (8)$$

This equation will be used to determine the motion of a driven pendulum. For the case of an undriven pendulum, $\zeta = 0$. In this case Eq. 11 becomes:

$$\ddot{\phi} = -\sin(\phi) \quad (9)$$

This equation will be used to determine the motion of a simple, undriven pendulum. We will now solve and plot these equations to better understand the motion of these pendulums in Sec. 3.

3 Results

To better understand the motion of these pendulums, we wrote Python code (found in Sec.A) to solve the differential equations(Eq. 9 and Eq. 11) for a range of initial conditions and angular frequencies. Using this code, we generated plots of the solutions, providing a way of understanding each pendulums motion.

Fig. 1 shows angular displacement $\phi(t)$ (in radians) plotted against time t in seconds for an undriven pendulum described by Eq. 9 for initial angular displacements: $\phi_0 = 0.5, 1.5, 2, 3rad$. The pendulum is released from rest in every case, meaning $\dot{\phi}_0 = 0rad/s$ (initial angular velocity is 0 radians per second).

We note that as the initial angular displacement, ϕ_0 , increases, both the amplitude and period of the oscillation increase. This means that as our initial angular displacement increases each swing of the pendulum takes longer and reaches a greater angular displacement at the end of each swing. For small angular displacements, the pendulum's motion is approximately simple harmonic motion(a result taken from [5]). In this case, the amplitude of the oscillation is ϕ_0 , and

the period, T , is given by $T = 2\pi\sqrt{\frac{L}{g}}$. As ϕ_0 increases, the approximation becomes less accurate, causing the period to change from this simple harmonic motion approximation. The amplitude remains equal to the initial displacement ϕ_0 (due to conservation of energy in the system see [2]), and the period becomes greater with larger initial angles. This relationship between ϕ_0 and the period is caused by the decrease in the accuracy of the simple harmonic motion approximation at greater initial angular displacements a more complete study is found in [5].

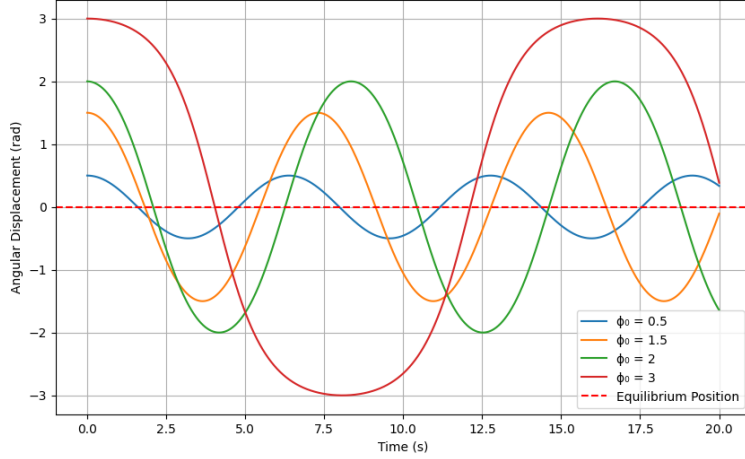


Figure 1: Angular displacement $\phi(t)$ (in radians) plotted against time t in seconds for an undriven pendulum governed by $\ddot{\phi} = -\sin(\phi)$ (equation of motion for a driven pendulum). The pendulum is released with zero initial angular velocity ($\dot{\phi} = 0 \text{ rad/s}$) and varying initial angular displacements: $\phi(0) = 0.5 \text{ rad}$ (blue), $\phi(0) = 1.5 \text{ rad}$ (orange), $\phi(0) = 2 \text{ rad}$ (green) and $\phi(0) = 3 \text{ rad}$ (red). The equilibrium position (ϕ_0) is indicated by the dashed red line. The path of motion in most cases is oscillatory (in some cases it is approximately simple harmonic motion), with amplitudes and periods increasing as ϕ_0 increases.

Fig. 2 shows the angular displacement of a driven pendulum over time, as described by Eq. 11, for various angular frequencies (ω), and initial angular displacements (ϕ_0). Each plot has a different angular frequency ω (5, 10, 15, 20, 25 or 30 rad/s). For every plot the pendulum is released from an angular displacement ϕ_0 (0.5 rad (blue), 1.5 rad (yellow), 2 rad (green) and 3 rad (red)), and an initial angular velocity $\dot{\phi}(0) = 0 \text{ rad/s}$.

Our plot shows that for most values of ω and ϕ_0 we produce stable oscillations, with exception to the case $\omega = 5$ and $\phi_0 = 3 \text{ rad}$. In this case the pendulum motion is not stable meaning the pendulum is not periodic and shows apparent randomness in its oscillations. This is caused by the additional driving factor, gravity and initial conditions. It should be noted in this case ω was our smallest tested value 5 rad/s and $\phi_0 = 3 \text{ rad}$ was our largest tested initial angular displacement. Given this it seems small ω and large ϕ_0 are potential factors in creating unstable pendulum motion. The stable pendulum motions are oscillations which have smaller high-frequency and low-amplitude oscillations within the bigger waves. The cause of these small oscillations is the driving factor, hence as when the angular frequency (ω) is increased we see a higher frequency of the low-amplitude oscillations.

It should be noted that the equilibrium position about which the pendulum oscillates shifts depending on the conditions. Majority of cases show oscillations about $\phi = 0$, as is the case for simple pendulum motion, some cases show oscillations about $\phi = \pi$, indicating that we produce an inverted pendulum. This suggests that a sufficiently high angular frequency in combination with a large enough initial angular displacement we can stabilize inverted pendulum motion. We will now preform a more complete study of the inverted pendulum in Sec. 3.1.

3.1 Stability of the inverted pendulum

We are now lead to study the conditions leading to the stability of the inverted pendulum motions observed in Fig. 2. We will do this by using a slight adaption of the results found in section 3 of

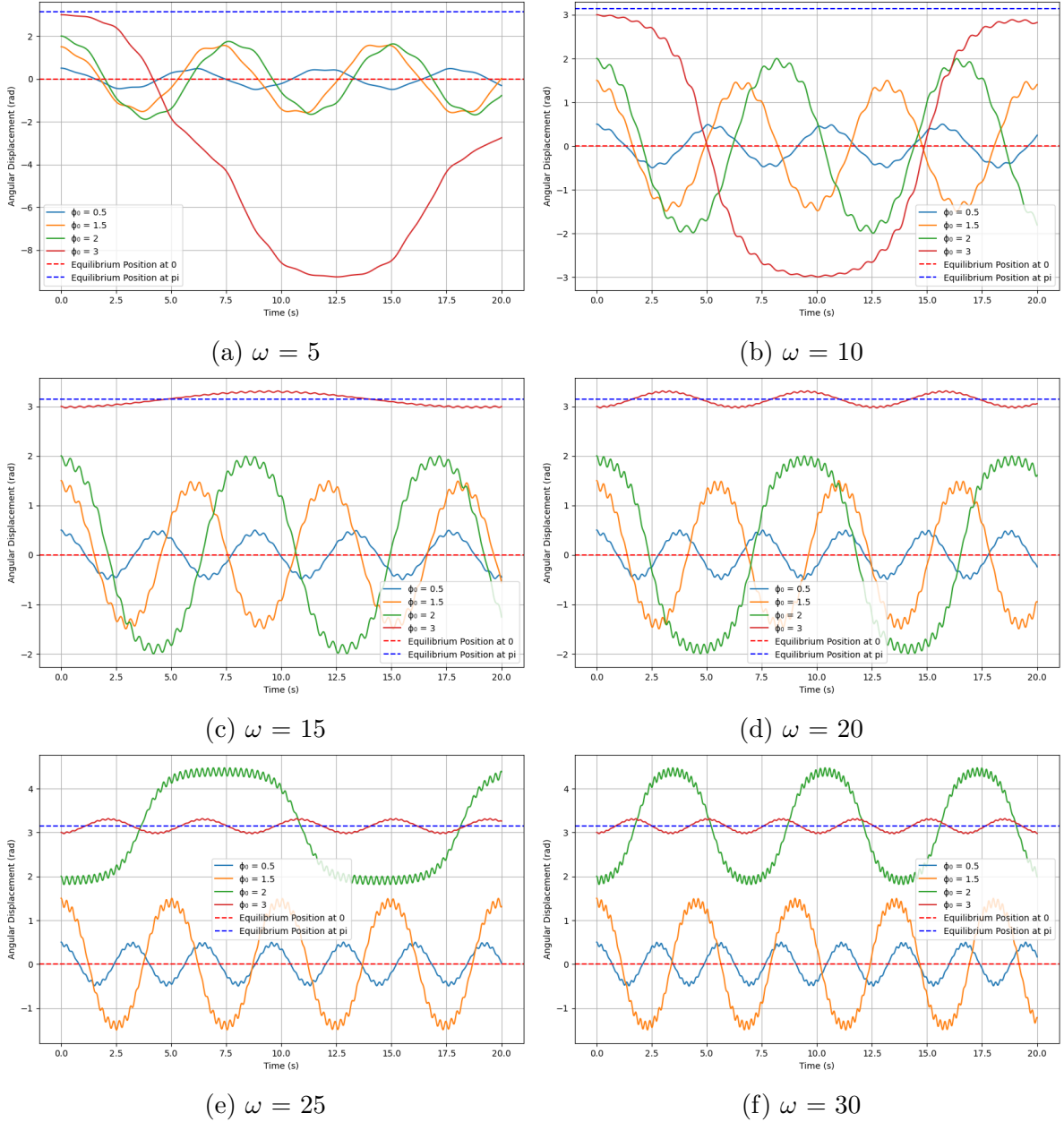


Figure 2: Angular displacement ($\phi(t)$ in radians against of a driven pendulum governed by $\ddot{\phi} = -(1 + \zeta\omega^2\cos(\omega t))\sin(\phi)$, plotted against time t (in seconds). A small driving factor is used ($\zeta = 0.1$) with angular frequency values of $\omega = 5, 10, 15, 20, 25$ and 30 (separately shown in each figure). In all cases, the pendulum is released with zero initial angular velocity ($\dot{\phi} = 0$) and varying initial angular displacements: $\phi_0(0.5\text{rad}(\text{blue}), 1.5\text{rad}(\text{yellow}), 2\text{rad}(\text{green})$ and $3\text{rad}(\text{red}))$. The equilibrium positions at $\phi = 0$ (red dashed line) and $\phi = \pi$ (blue dashed line) are included to show the points about which oscillations can occur. The motion changes depending on the angular frequency and initial conditions. We see that in general within each plot the oscillations with high-amplitude and low-frequency have amplitudes and periods increasing with ϕ_0 , and as the angular frequency increases we see the oscillations with low-amplitude and high-frequency experience a growing frequency.

[3]. To do this, we consider an approximate solution to Eq. 11, which describes the pendulum's motion given by:

$$\ddot{\phi}(t) = -(1 + \zeta\omega^2 \cos(\omega t)) \sin(\phi(t)). \quad (10)$$

For convenience, we rewrite this as:

$$\ddot{\phi}(t) + \sin(\phi(t)) + \zeta\omega^2 \sin(\phi(t)) \cos(\omega t) = 0. \quad (11)$$

The plots in Figure 2 have suggested a solution of the form:

$$\phi(t) = \theta(t) + \delta(t), \quad (12)$$

where $\delta(t)$ represents a high-frequency, low-amplitude function, and $\theta(t)$ represents a low-frequency, high-amplitude function. This solution is suggested by the apparent superposition of two waves observed in the plot. Applying a first-order Taylor series approximation, we obtain:

$$\sin(\phi(t)) = \sin(\theta(t) + \delta(t)) \approx \sin(\theta(t)) + \delta(t) \cos(\theta(t)). \quad (13)$$

We consider the motion averaged over the period T of the high-frequency oscillations of $\delta(t)$. In this case, we can regard $\theta(t)$ as approximately constant $\theta(t) \approx \theta$. Using these approximations as tools, upon further work(found in [3]) we arrive at the approximate condition for pendulum stability:

$$\zeta\omega^2 > 2g \quad \text{or} \quad \zeta\omega > \sqrt{2} \quad \text{or} \quad \zeta > \sqrt{\frac{2}{\omega^2}} \quad (14)$$

It should be noted these approximations are only valid when $\zeta \ll 1$, and $\omega \gg 1$. Both of these hold true for the pendulum motion we plotted($\zeta = 0.1$, $\omega > 5$).

We will now test how this condition connects with our plotted results. Recall that in our plots, we set $\zeta = 0.1$ and varied $\omega = 5, 10, 15, 20, 25$ and 30rad/s . For $\omega = 5\text{rad/s}$, we did not observe stable inverted pendulums. In this case, $\zeta\omega^2 = 2.5 < 2g$, $\zeta\omega = 0.5 < \sqrt{2}$ and $\zeta = 0.1 < \sqrt{\frac{2}{\omega^2}}$, all of which indicate instability, as predicted by our condition. Similarly, for $\omega = 10\text{rad/s}$, the inequalities $\zeta\omega^2 = 10 < 2g$, $\zeta\omega = 1 < \sqrt{2}$ and $\zeta = 0.1 < \sqrt{\frac{2}{\omega^2}}$ again predict instability, consistent with our observations. In the cases where $\omega = 15, 20, 25$ and 30rad/s we have $\zeta\omega^2 \geq 22.5 > 2g$ in all cases. This suggests the possibility of stable inverted pendulum motion, which is what we observed in the plots, and so we have a working condition to predict whether our environment can produce a stable inverted pendulum.

4 Conclusions

In this report, we derived the equations of motion for both driven and undriven pendulums, and solved and visualized the equations using Python. Our results investigated a variety of these results such as the stability of the inverted pendulum under specific conditions, and how angular frequency and initial angular displacement impact pendulum motion. We did not consider the effects of damping, which would change the equations of motion. Adding damping as a factor to the pendulum motion, improving stability condition approximations, and considering other additional variations is the next step forward with this report. Further work has been done on these topics and can be found in [3]. We should also consider the lack of consideration of the chaotic motion that was found. Further work on the topic is found here [4].

References

- [1] G. Gribakin, *MTH2021 Classical Mechanics, Lecture notes* (Queen's University Belfast, 2024).
- [2] Herbert Goldstein, *Classical Mechanics*, 2nd ed. (Addison-Wesley, Harvard, 1964).
- [3] E. I. Butikov, An improved criterion for Kapitza's pendulum stability, *J. Phys. A: Math. Theor.* **44**, 295202 (2011).

- [4] *The Road to Chaos* (accessed 09/12/24).
 [5] *The Simple Pendulum* (accessed 02/12/24).

A Codes

The Python code listed below was used to produce the graphs of $\phi(t)$, Eq. (9) shown in Fig. 1 and Eq. (11) shown in Fig. 2. The code is an application of the packages numpy, odeint and matplotlib.pyplot.

```

1  # -*- coding: utf-8 -*-
2  """MTH2031
3
4  Automatically generated by Colab.
5
6  Original file is located at
7      https://colab.research.google.com/drive/1bPMHlyUVCsZ_sG6d49rg37ngRNx-wUmv
8  """
9
10 ## Importing relevant packages
11 import numpy as np
12 from scipy.integrate import odeint
13 import matplotlib.pyplot as plt
14
15 # Undriven Pendulum (Eq.~11):
16
17 # This function models the undriven pendulum
18 def pendulum_equation3(vec, t):
19     phi, phi_dot = vec # vec contains angular displacement and angular
20     velocity
21     # Angular velocity
22     dphi_dt = phi_dot
23     # Angular acceleration
24     dphi_dot_dt = -np.sin(phi)
25     return [dphi_dt, dphi_dot_dt]
26
27 # Array for initial conditions
28 phi_0 = [[0.5, 0], [1.5, 0], [2, 0], [3, 0]]
29 # Time array
30 t = np.linspace(0, 20, 10000) # from t=0 to t=20 with 1000 points
31 # Solve the equation for each initial condition
32 solutions = []
33 for i in phi_0:
34     solution = odeint(pendulum_equation3, i, t)
35     solutions.append(solution)
36
37 # Plotting the solutions
38 plt.figure(figsize=(10, 6)) # adjusting figure size
39 for i, solution in enumerate(solutions):
40     plt.plot(t, solution[:, 0], label=f' = {phi_0[i][0]}') # Plot (t)
41     for each initial condition, labeling done using f string feature.
42
43 plt.axhline(y=0, color='r', linestyle='--', label='Equilibrium Position') #
44     Add label equilibrium line
45
46 plt.xlabel('Time (s)') # X-axis label
47 plt.ylabel('Angular Displacement (rad)') # Y-axis label
48 plt.legend() # Add legend to identify curves
49 plt.grid(True) # Add grid
50 plt.show() # Display the plot
51
52 # Driven Pendulum (Equation 10):
53
54 # This function models the driven pendulum
55 def pendulum_equation4(vec, t, zeta, omega):

```

```

49     phi, phi_dot = vec # Angular displacement and angular velocity
50     dphi_dt = phi_dot # Angular velocity
51     dphi_dot_dt = (1 + zeta * (omega**2) * np.cos(omega * t)) *-1* np.sin(phi)
    # Angular acceleration with driving term
52     return [dphi_dt, dphi_dot_dt]
53 phi_0 = [[0.5, 0], [1.5, 0], [2, 0], [3, 0]] # Initial conditions
54 omega_values = [5, 10, 15, 20, 25, 30] # Angular frequencies
55 zeta = 0.1 #setting zeta to a low value(0.1)
56 # Solve the equation for each omega and initial condition
57 for omega in omega_values:
58     plt.figure(figsize=(10, 6))#adjusting figure size
59     for initial_state in phi_0:
60         # Solve the differential equation
61         solution = odeint(pendulum_equation4, initial_state, t, args=(zeta,
        omega))
62         plt.plot(t, solution[:, 0], label=f'          = {initial_state[0]}') #
        Producing the plot, labeling done using f string feature
63         solution = []
64         plt.axhline(y=0, color='r', linestyle='--', label='Equilibrium Position at
        0') # Add and plot equilibrium line at 0
65         plt.axhline(y=np.pi, color='b', linestyle='--', label='Equilibrium
        Position at pi') # Add and plot equilibrium line at pi
66         plt.xlabel('Time (s)') # X-axis label
67         plt.ylabel('Angular Displacement (rad)') # Y-axis label
68         plt.legend() # Add legend to identify curves
69         plt.grid(True)#Showing grid for clarity
70         plt.show() # Display the plot

```